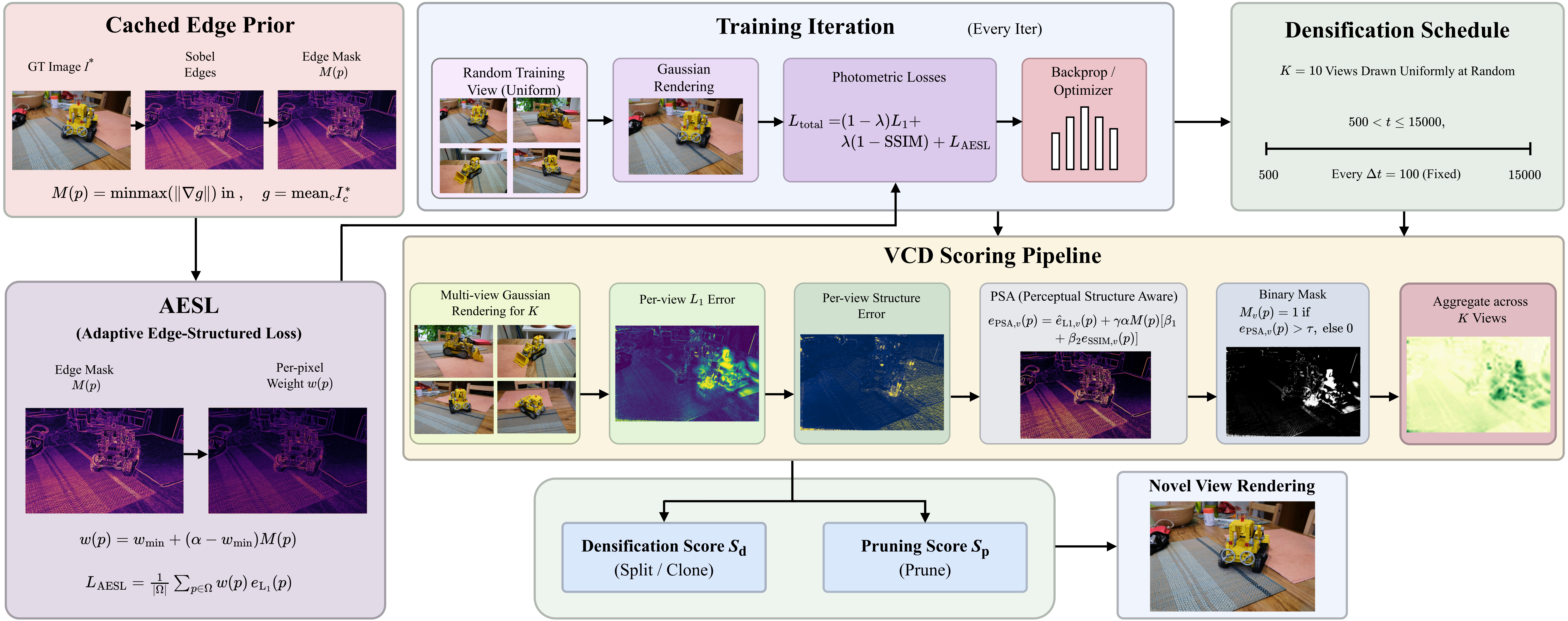

Cached Edge Prior
GT Image I^*
Sobel Edges
Edge Mask $M(p)$
$M(p) = \text{minmax}(\|\nabla g\|)$ in , $g = \text{mean}_c I_c^*$
Training Iteration
(Every Iter)
Random Training View (Uniform)
Gaussian Rendering
Photometric Losses
$L_{\text{total}} = (1-\lambda)L_1 + \lambda(1-\text{SSIM}) + L_{\text{AESL}}$
Backprop / Optimizer
Densification Schedule
$K = 10$ Views Drawn Uniformly at Random
$500 < t \leq 15000,$
500
Every $\Delta t = 100$ (Fixed)
15000
AESL
(Adaptive Edge-Structured Loss)
Edge Mask $M(p)$
Per-pixel Weight $w(p)$
$w(p) = w_{\min} + (\alpha - w_{\min})M(p)$
$L_{\text{AESL}} = \frac{1}{|\Omega|}\sum_{p\in\Omega} w(p)\, e_{\text{L}_1}(p)$
VCD Scoring Pipeline
Multi-view Gaussian Rendering for K
Per-view L_1 Error
Per-view Structure Error
PSA (Perceptual Structure Aware)
$e_{\text{PSA},v}(p) = \hat{e}_{\text{L1},v}(p) + \gamma\alpha M(p)[\beta_1 + \beta_2 e_{\text{SSIM},v}(p)]$
Binary Mask
$M_v(p) = 1$ if $e_{\text{PSA},v}(p) > \tau,$ else 0
Aggregate across K Views
Densification Score S_{d}
(Split / Clone)
Pruning Score S_{p}
(Prune)
Novel View Rendering